

PKP DESIGN RULE MANUAL

PDK Release 1.1

List of tables

TABLE1.2.1 RULE TYPES AND THEIR DESCRIPTION.	3
TABLE2.1.1 FRONT-END-OF-LINE LAYER INFORMATION.....	5
TABLE2.2.1 MIDDLE-OF-LINE LAYER INFORMATION.....	5
TABLE2.3.1 BACK-END-OF-LINE LAYER INFORMATION.....	6
TABLE2.4.1 MARKER LAYER INFORMATION.....	6
TABLE3.1.1 DESIGN REQUIREMENTS.....	7
TABLE3.2.1 ACT DESIGN RULES.	8
TABLE3.3.1 GATE DESIGN RULES.	10
TABLE3.4.1 NS DESIGN RULES.	12
TABLE3.5.1 NW DESIGN RULES.....	13
TABLE3.6.1 NIM/PIM/VT DESIGN RULES.....	14
TABLE3.7.1 CT DESIGN RULES.....	15
TABLE3.8.1 GCUT DESIGN RULES.....	17
TABLE3.9.1 CT_PR DESIGN RULES.	19
TABLE3.10.1 VCT DESIGN RULES.	21
TABLE3.11.1 VCT_PR DESIGN RULES.....	23
TABLE3.12.1 VCG DESIGN RULES.....	25
TABLE3.13.1 M0 DESIGN RULES.....	27
TABLE3.14.1 M0_PR DESIGN RULES.	28
TABLE3.15.1 V0 DESIGN RULES.....	28

TABLE3.16.1 M1 DESIGN RULES.....	30
TABLE3.17.1 V1 DESIGN RULES.....	32
TABLE3.18.1 M2 DESIGN RULES.....	34
TABLE3.19.1 V2 DESIGN RULES.....	34
TABLE3.20.1 M3 DESIGN RULES.....	34
TABLE3.21.1 V3 DESIGN RULES.....	35
TABLE3.22.1 M4 DESIGN RULES.....	35
TABLE3.23.1 V4 DESIGN RULES.....	35
TABLE3.24.1 M5 DESIGN RULES.....	36
TABLE3.25.1 V5 DESIGN RULES.....	36
TABLE3.26.1 M6 DESIGN RULES.....	36
TABLE3.27.1 V6 DESIGN RULES.....	37
TABLE3.28.1 M7 DESIGN RULES.....	37
TABLE3.29.1 V7 DESIGN RULES.....	37
TABLE3.30.1 M8 DESIGN RULES.....	38
TABLE3.31.1 V8 DESIGN RULES.....	38
TABLE3.32.1 M9 DESIGN RULES.....	38
TABLE3.33.1 V9 DESIGN RULES.....	39
TABLE3.34.1 M10 DESIGN RULES.....	39
TABLE3.35.1 V10 DESIGN RULES.....	39
TABLE3.36.1 M11 DESIGN RULES.....	40
TABLE3.37.1 V11 DESIGN RULES.....	40

TABLE3.38.1 M12 DESIGN RULES.....40

TABLE3.39.1 V12 DESIGN RULES.....41

TABLE3.40.1 M13 DESIGN RULES.....41

List of Figures

FIG. 1.2.1 ILLUSTRATION OF RULE TYPES.....	4
FIG. 3.1.1 ILLUSTRATION OF DESIGN REQUIREMENTS.	7
FIG. 3.2.1 ILLUSTRATION OF ACT DESIGN RULES.....	9
FIG. 3.3.1 ILLUSTRATION OF GATE DESIGN RULES.....	11
FIG. 3.4.1 ILLUSTRATION OF NS DESIGN RULES.....	12
FIG. 3.5.1 ILLUSTRATION OF NW DESIGN RULES.	13
FIG. 3.6.1 ILLUSTRATION OF NIM/PIM/VT DESIGN RULES.	14
FIG. 3.7.1 ILLUSTRATION OF CT DESIGN RULES.	16
FIG. 3.8.1 ILLUSTRATION OF GCUT DESIGN RULES.	18
FIG. 3.9.1 ILLUSTRATION OF CT_PR DESIGN RULES.	20
FIG. 3.10.1 ILLUSTRATION OF VCT DESIGN RULES.....	22
FIG. 3.11.1 ILLUSTRATION OF VCT_PR DESIGN RULES.	24
FIG. 3.12.1 ILLUSTRATION OF VCG DESIGN RULES.	26
FIG. 3.13.1 ILLUSTRATION OF M0 DESIGN RULES.	27
FIG. 3.15.1 ILLUSTRATION OF V0 DESIGN RULES.....	29
FIG. 3.16.1 ILLUSTRATION OF M1 DESIGN RULES.	31
FIG. 3.17.1 ILLUSTRATION OF V1 DESIGN RULES.....	33

Table of Contents

1 CONCERNING THIS MANUAL	1
1.1 License	1
1.2 Terminology and Conventions Used in This Manual	2
1.2.1 Terminology	2
1.2.2 Notes On Conventions	2
1.2.3 Rule Name Convention.....	2
2 LAYER INFORMATION	5
2.1 Front-End-of-Line (FEOL) Layer Information	5
2.2 Middle-of-Line (MOL) Layer Information.....	5
2.3 Back-End-of-Line (BEOL) Layer Information	6
2.4 Marker Layer Information.....	6
3 PHYSICAL DESIGN RULES	7
3.1 Design requirements.....	7
3.2 ACT design rules.....	8
3.3 GATE design rules.....	10
3.4 NS design rules.....	12
3.5 NW design rules.....	13
3.6 NIM/PIM/VT design rules	14
3.7 CT design rules	15

3.8 GCUT design rules	17
3.9 CT_PR design rules.....	19
3.10 VCT design rules	21
3.11 VCT_PR design rules	23
3.12 VCG design rules.....	25
3.13 M0 design rules.....	27
3.14 M0_PR design rules.....	28
3.15 V0 design rules.....	28
3.16 M1 design rules.....	30
3.17 V1 design rules.....	32
3.18 M2 design rules.....	34
3.19 V2 design rules.....	34
3.20 M3 design rules.....	34
3.21 V3 design rules.....	35
3.22 M4 design rules.....	35
3.23 V4 design rules.....	35
3.24 M5 design rules.....	36
3.25 V5 design rules.....	36
3.26 M6 design rules.....	36
3.27 V6 design rules.....	37
3.28 M7 design rules.....	37
3.29 V7 design rules.....	37

3.30 M8 design rules.....	38
3.31 V8 design rules.....	38
3.32 M9 design rules.....	38
3.33 V9 design rules.....	39
3.34 M10 design rules.....	39
3.35 V10 design rules.....	39
3.36 M11 design rules.....	40
3.37 V11 design rules.....	40
3.38 M12 design rules.....	40
3.39 V12 design rules.....	41
3.40 M13 design rules.....	41

1 Concerning This Manual

1.1 License

PKP PDK™ Predictive Process Design Kit

BSD 3-Clause License

Copyright 2025 Lining Zhang

Redistribution and use in source and binary forms, with or without modification, are permitted provided that the following conditions are met:

1.Redistributions of source code must retain the above copyright notice, this list of conditions and the following disclaimer.

2.Redistributions in binary form must reproduce the above copyright notice, this list of conditions and the following disclaimer in the documentation and/or other materials provided with the distribution.

3.Neither the name of the copyright holder nor the names of its contributors may be used to endorse or promote products derived from this software without specific prior written permission.

THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS" AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT HOLDER OR CONTRIBUTORS BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT2(INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

1.2 Terminology and Conventions Used in This Manual

1.2.1 Terminology

BEOL:Refers to ‘back-end-of-line’ interconnect (M1-M13) and via layers (V1-V12).

FEOL:Refers to ‘front-end-of-line’ layers, viz. WELL, ACT, NSH,GATE, GCUT,

SDB,NIM,PIM,SVT_N,SVT_P,LVT_N,LVT_P,SLVT_N,SLVT_P.

MOL:Refers to ‘middle-of-line’ layers, viz. CT, CT_PR, VCT,VCT_PR,VCG.

1.2.2 Notes On Conventions

The figures provided in this manual are not drawn to scale and in some figures the geometries have been accentuated for the purpose of illustration. Therefore, designers must not to base their interpretation of rules solely on the scale at which the figures are drawn.

1.2.3 Rule Name Convention

Rule name for intra-layer rules: PrimaryLayer.Type.Number

Rule name for inter-layer rules: PrimaryLayer.SecondaryLayer.Type.Number

An inter-layer rule refers to a rule defined between two different layers and an intra-layer rule refers to a rule defined for a single layer.

‘PrimaryLayer’ and ‘SecondaryLayer’ represent the layers for which a rule is written. The string ‘Type’ may have the values listed in Table:

Abbreviation	Type	Description	Figure
A	Area	minima area rule of (inter/intra layer).	Fig.1.2.1(a)
L	Length	A length (intra layer) rule. Denotes the length of a polygon's edge. The scope of this rule type is not limited to just the distance between the interior side of edges of a layer geometry.	Fig.1.2.1(b)
W	Width	A width rule (intra layer). Denotes the distance between interior side of edges of a specific layer.	Fig.1.2.1(c)
S	Spacing	A spacing rule (inter/intra layer). Denotes the distance between the exterior side of edges of one or two layers.	Fig.1.2.1(d)

EN	Enclosure	An enclosure rule (inter layer). Denotes the distance between the exterior side of one layer's edge and the interior side of another layer's edge, where one of the layer geometry lies completely within another layer geometry	Fig.1.2.1(e)
EX	Extension	An extension rule (inter layer). Denotes the distance between the exterior side of one layer's edge and the interior side of another layer's edge, where one of the layer geometry lies partially within another layer geometry	Fig.1.2.1(f)
O	Overlap	An overlap rule (inter layer). Denotes the distance between interior side of edges of two layers.	Fig.1.2.1(g)
D	Density	Density Rule	
DFM	DFM	DFM Rule	
GEOMETRY		A geometry check. Stipulates the permitted shape geometry.	
LUP	LatchUP	Latch-up prevention rule.	
ANT	Antenna	Antenna Rule check.	
R	Recommend	Recommended Rule(may be waived)	
GR[NC]	Guideline	Guideline Rule (may be waived), the ones that do not fall under any of the above categories ([NC]rule not be check)	
SRAM	SRAM	SRAM related Rule, only apply to SRAM cells	
AUX	Auxiliary	the ones that do not fall under any of the above categories (may not be waived)	
Dev	Device	Device rule	

Table1.2.1 Rule types and their description.

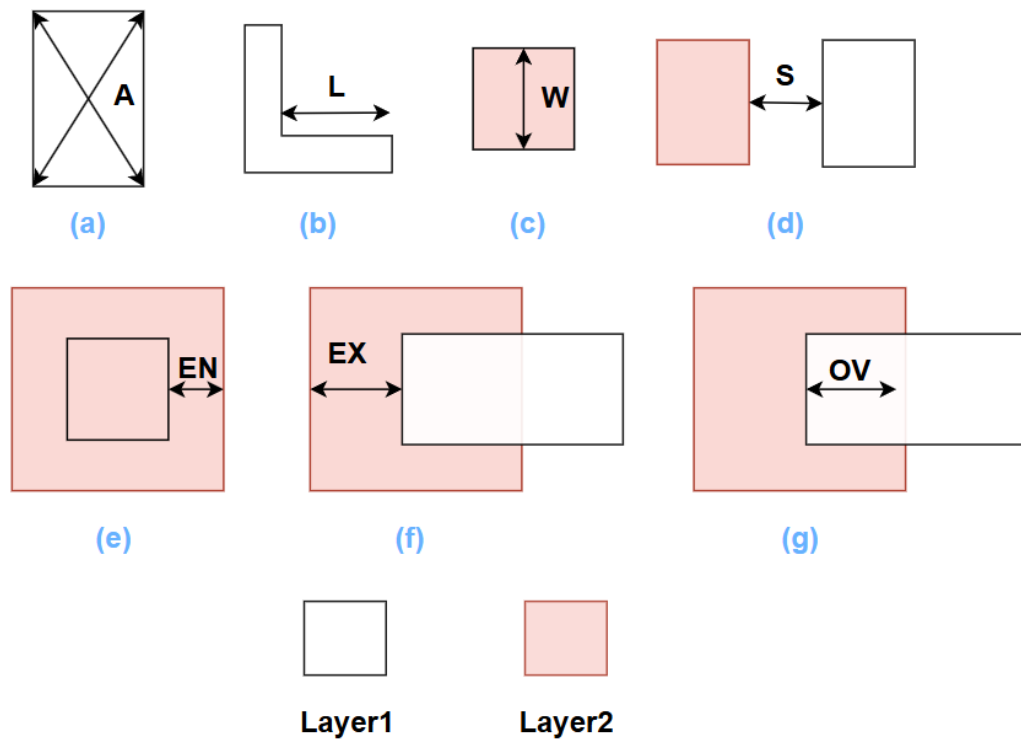


Fig. 1.2.1 Illustration of rule types.

2 Layer Information

2.1 Front-End-of-Line (FEOL) Layer Information

Layer Name	Mask ID	Description
NW	1	N-Well
ACT	2	ACTIVE
NSH	3	NANOSHEET
GATE	4	Metal Gate
GCUT	5	Gate metal Cut
SDB	6	Single Diffusion Break
NIM	7	N-Implant
PIM	8	P-Implant
SVT_N		
SVT_P		
LVT_N		
LVT_P		
SLVT_N		
SLVT_P		

Table2.1.1 Front-end-of-line layer information.

2.2 Middle-of-Line (MOL) Layer Information

Layer Name	Mask ID	Description
CT	9	Source or Drain interconnect layer
CT_PR	10	CT power rail
VCT	11	VCT, connects CT to M0
VCT_PR	12	VCT power rail
VCG	13	Gate interconnect layer

Table2.2.1 Middle-of-line layer information.

2.3 Back-End-of-Line (BEOL) Layer Information

Layer Name	Mask ID	Description
M0	14	M0
M0_PR	15	M0 power rail
V0	16	Via0, connects M0 to M1
M1	17	M1
V1	18	Via1
M2	19	M2
V2	20	Via2
M3	21	M3
V3	22	Via3
M4	23	M4
V4	24	Via4
M5	25	M5
V5	26	Via5
M6	27	M6
V6	28	Via6
M7	29	M7
V7	30	Via7
M8	31	M8
V8	32	Via8
M9	33	M9
V9	34	Via9
M10	35	M10
V10	36	Via10
M11	37	M11
V11	38	Via11
M12	39	M12
V12	40	Via12
M13	41	M13

Table2.3.1 Back-end-of-line layer information.

2.4 Marker Layer Information

Layer Name	Mask ID	Description
boundary		boundary

Table2.4.1 Marker layer information.

3 Physical Design Rules

3.1 Design requirements

Rule number	Description
GR_1	All the geometry design must be an integer multiple of 0.001um
GR_2	must be on an interger multiple of 0.001um
GR_3	only shapes of geometry that are orthogonal or 45-degree angle are allowed
GR_4	shapes with acute angles between line segments are not allowed
GR_5[NC]	only unidirectional, straight transistor channels are allowed
GR_6[NC]	Recommended to design simple rectangular shape geometry as possible,avoid L,U,O shapes
GR_7[NC]	all line-end must be rectangular
GR_8[NC]	Self-intersection shape are not allowed
GR_9[NC]	Design shape must be polygons
GR_10[NC]	Make sure the designs are DRC clean

Table3.1.1 Design requirements.

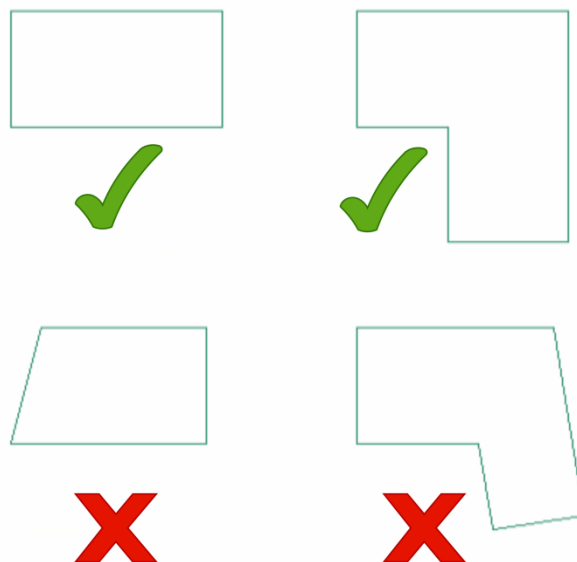


Fig. 3.1.1 Illustration of Design requirements.

3.2 ACT design rules

Rule number	Description	Opt.	Design value	Unit
ACT_W_1	minimum vertical width of ACT (outside the layer SRAMMRK)	>=	15	nm
ACT_W_2_SRAM	minimum vertical width of ACT (interacting with the layer SRAMMRK)	>=	10	nm
ACT_W_3	minimum horizontal width of ACT	>=	15	nm
ACT_W_4	width in GATE poly direction	<=	50	nm
ACT_S_1	Minimum vertical spacing of ACT (outside the layer SRAMMRK)	>=	60	nm
ACT_S_2_SRAM	Minimum vertical spacing of ACT (interacting with the layer SRAMMRK)	>=	40	nm
ACT_S_2A	Minimum horizontal spacing between ACT layers forming source/drain regions (of different transistors) on different nets	>=	48	nm
ACT_S_2B	Minimum horizontal spacing of ACT	>=	16	nm
ACT_S_3B	Space between NW and ACT in GATE poly direction (outside the layer SRAMMRK)	>=	27	nm
ACT_S_4B	Space between NW and ACT in GATE poly direction (interacting with the layer SRAMMRK)	>=	13.5	nm
ACT_AUX_3	A notch in ACT along vertical axis is not allowed.			
ACT_A_1	ACT area (outside the layer SRAMMRK)	>=	225	nm ²
ACT_A_2_SRAM	ACT area (interacting with the layer SRAMMRK)	>=	150	nm ²

Table3.2.1 ACT design rules.

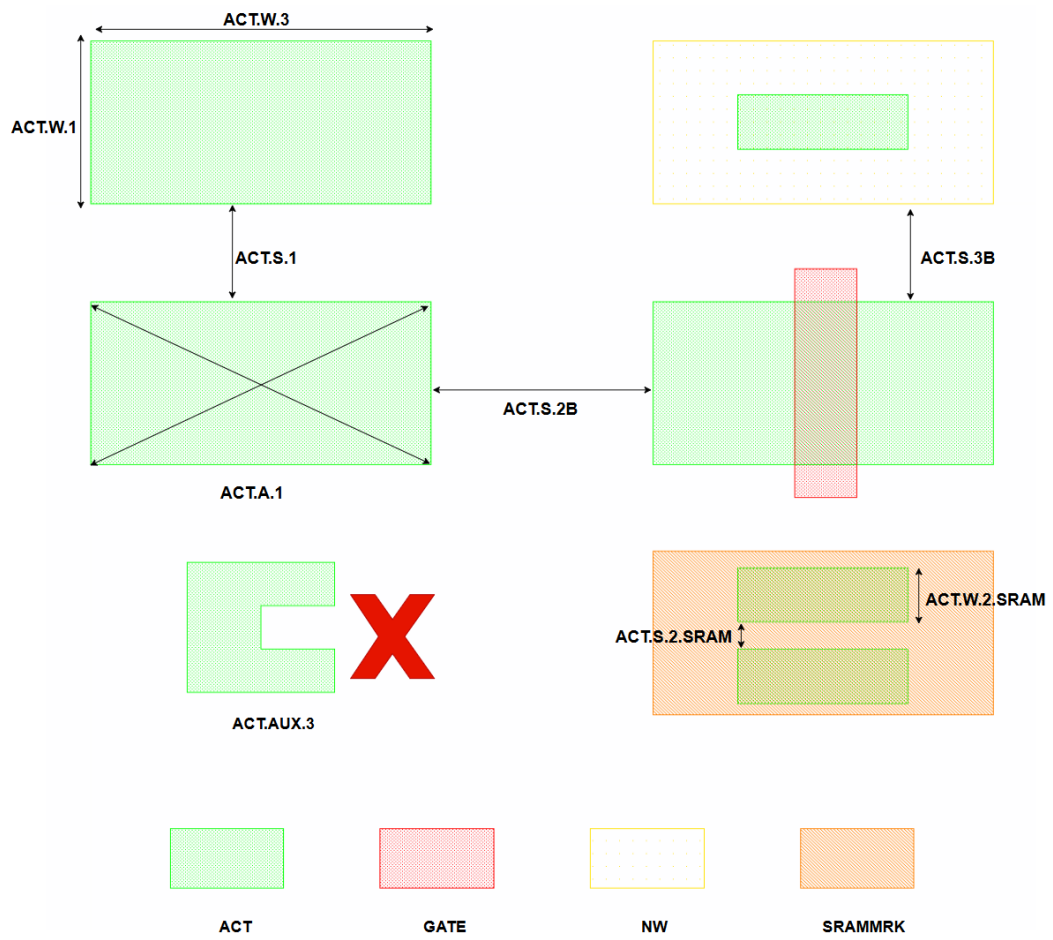


Fig. 3.2.1 Illustration of ACT design rules.

3.3 GATE design rules

Rule number	Description	Opt.	Design value	Unit
GATE_W_1	Exact horizontal GATE width (note: Lg)	>=	16	nm
GATE_W_2	Minimum vertical GATE width	>=	40	nm
GATE_S_1	Exact horizontal GATE pitch (not cut by GCUT and not interacting with the layer SRAMDRC)	==	48	nm
GATE_S_2	Minimum horizontal GATE spacing	>=	32	nm
GATE_AUX_1	GATE may not bend			
GATE_AUX_2	GATE may not be discontinuous along the vertical axis_ Use GCUT layer to mark cuts in the GATE			
GATE_AA_AUX_3	AA layer vertical edge may not lie inside, or coincide with, the GATE layer			
GATE_AA_EX_1	Minimum vertical extension of GATE (not cut by GCUT) past AA	>=	5	nm
GATE_AA_EX_2	Minimum horizontal extension of AA interacting with GATE (not cut by GCUT layer) past GATE (not cut by GCUT layer)	>=	16	nm
GATE_A_1	GATE area	>=	640	nm2

Table3.3.1 GATE design rules.

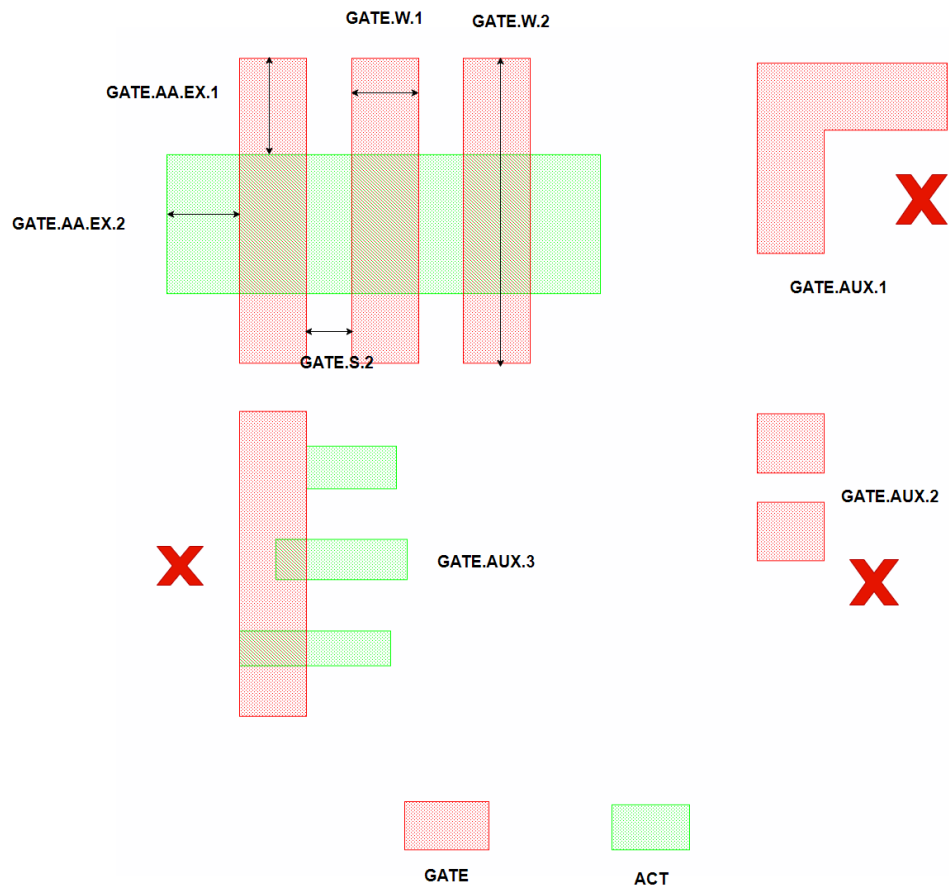


Fig. 3.3.1 Illustration of GATE design rules.

3.4 NS design rules

Rule number	Description	Opt.	Design value	Unit
NS_W_1	Vertical nanosheet width	$\geq$	10	nm
NS_W_2	Minimum horizontal nanosheet width	$\geq$	90	nm
NS_S_2	Exact vertical nanosheet spacing	$=$	20	nm
NS_AUX_1	nanosheet may not bend			

Table3.4.1 NS design rules.

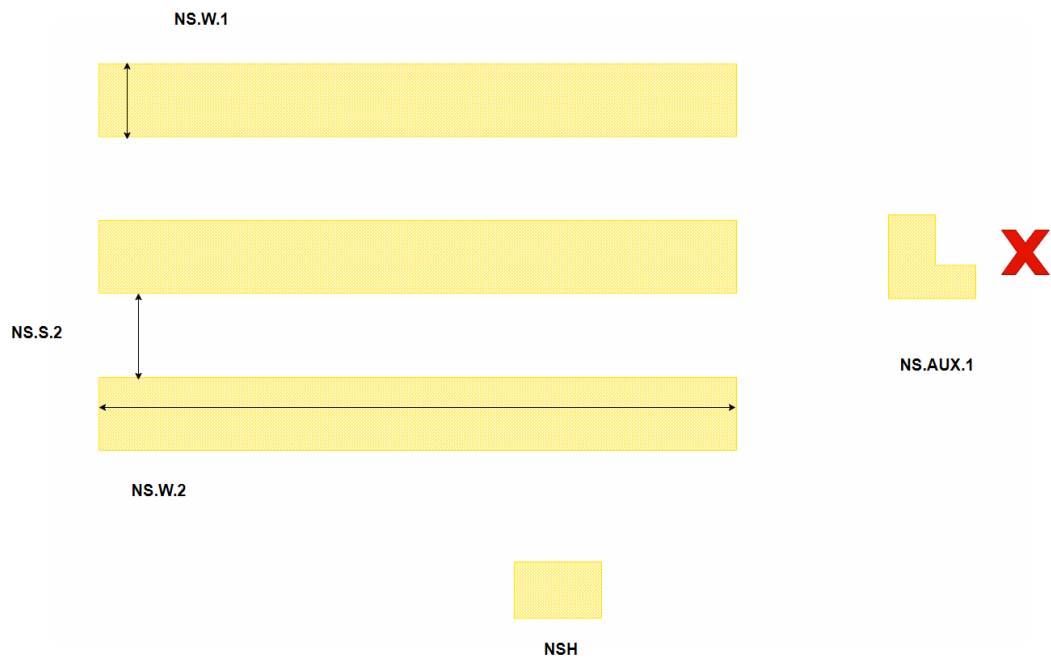


Fig. 3.4.1 Illustration of NS design rules.

3.5 NW design rules

Rule number	Description	Opt.	Design value	Unit
NW_W_1	Minimum horizontal width of WELL	>=	90	nm
NW_W_2	Minimum vertical width of WELL	>=	57	nm
NW_S_1	Minimum vertical spacing between WELL layer polygons	>=	90	nm
NW_S_2	Minimum horizontal spacing between WELL layer polygons	>=	45	nm
NW_A_1A	Minimum area of WELL	>=	5130	nm2
NW_A_1B	Minimum enclosed area of WELL	>=	5130	nm2
NW_GATE_EX_1	Minimum horizontal extension of WELL past GATE (not cut by GCUT layer)	>=	7	nm
NW_GATE_EX_2	Minimum vertical extension of WELL past GATE (not cut by GCUT layer) that does not interact with SRAMDRC	>=	7	nm

Table3.5.1 NW design rules.

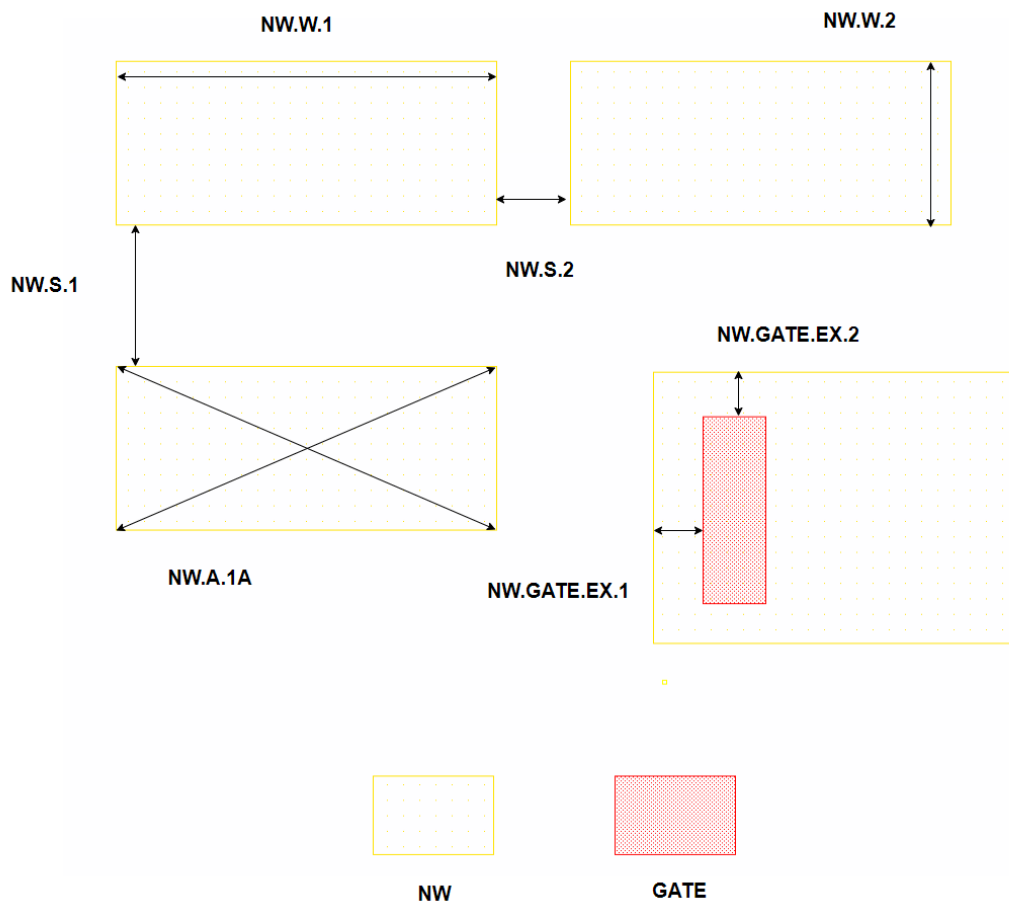


Fig. 3.5.1 Illustration of NW design rules.

3.6 NIM/PIM/VT design rules

Rule number	Description	Opt.	Design value	Unit
NIM_W_1	Minimum horizontal width of NIM	>=	48	nm
NIM_W_2	Minimum vertical width of NIM (outside the layer SRAMMRK)	>=	57	nm
NIM_W_3_SRAM	Minimum vertical width of NIM (interacting with the layer SRAMMRK)	>=	45	nm
NIM_S_1	NIM horizontal Space	>=	48	nm
NIM_S_2	horizontal Space between ACT and NIM	>=	48	nm
VT_AUX_2	VT layers (LVT, SLVT, and SRAMVT) may not overlap			
NIM.R_1	NIM and PIM may not overlap			
NIM_A_1	NIM area	>=	2736	nm2

Table3.6.1 NIM/PIM/VT design rules.

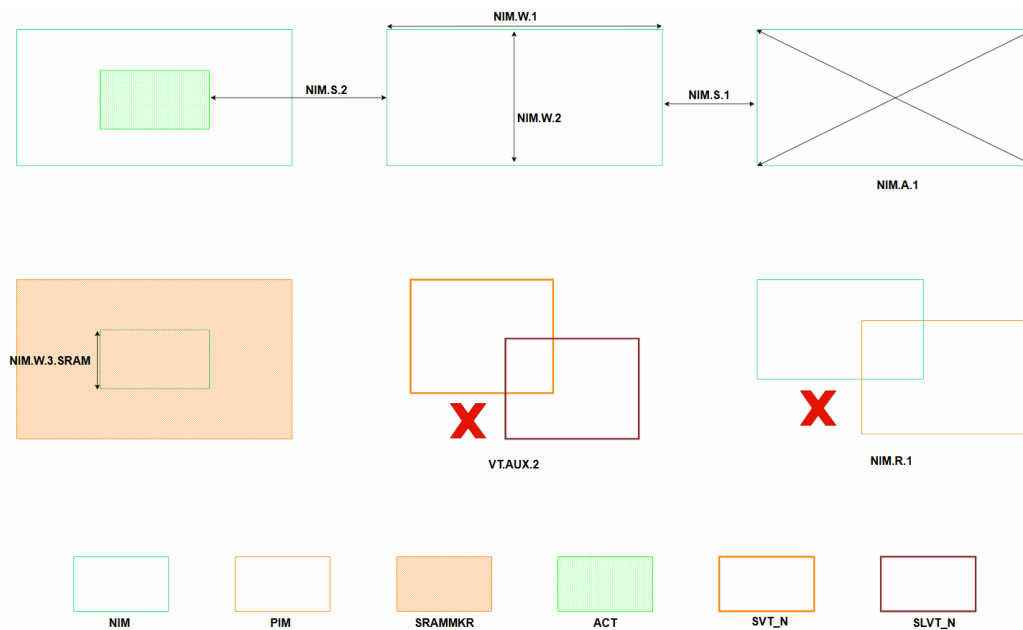


Fig. 3.6.1 Illustration of NIM/PIM/VT design rules.

3.7 CT design rules

Rule number	Description	Opt.	Design value	Unit
CT_W_1	Minimum horizontal width of CT	>=	10	nm
CT_W_2	Minimum vertical width of CT (outside the layer SRAMMRK)	>=	20	nm
CT_W_3_SRAM	Minimum vertical width of CT (interacting with the layer SRAMMRK)	>=	15	nm
CT_S_1	Minimum horizontal spacing between two CT layer polygons	>=	28	nm
CT_GATE_S_2	Minimum horizontal spacing between CT and GATE	>=	6	nm
CT_ACTIVE_OV_1	Minimum vertical overlap between CT and ACTIVE (outside the layer SRAMMRK)	>=	15	nm
CT_ACTIVE_OV_2_SRAM	Minimum vertical overlap between CT and ACTIVE (interacting with the layer SRAMMRK)	>=	10	nm
CT_VCT_OV_2	Minimum vertical overlap between CT and VCT	>=	15	nm
CT_GATE_AUX_1	CT and GATE may not overlap or touch			
CT_ACT_AUX_3	CT may not be entirely outside ACT			
CT_ACT_EX_1	Minimum vertical extension of CT past ACT	>=	8	nm

Table3.7.1 CT design rules.

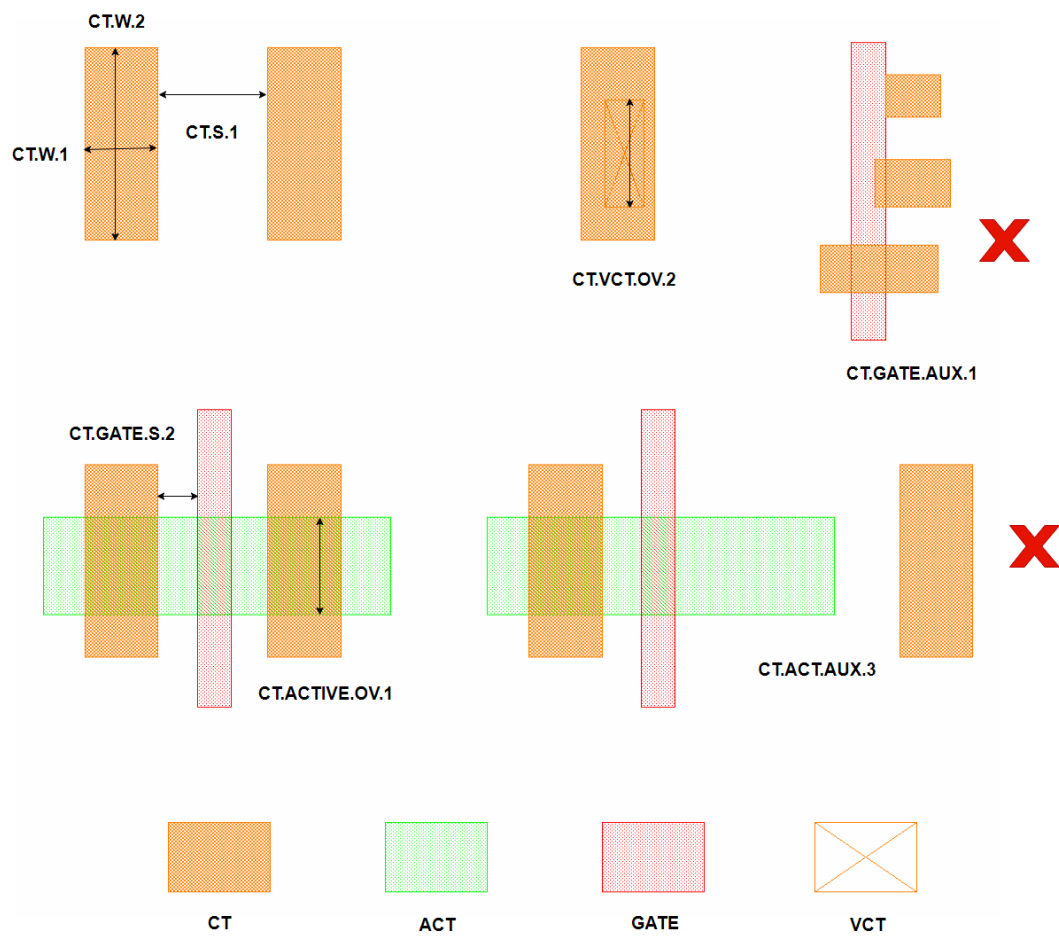


Fig. 3.7.1 Illustration of CT design rules.

3.8 GCUT design rules

Rule number	Description	Opt.	Design value	Unit
GCUT_W_1	Minimum vertical width of GCUT	$\geq$	20	nm
GCUT_W_2	Maximum vertical width	$\leq$	40	nm
GCUT_L_1	GCUT edge length in S/D direction	$\geq$	35	nm
GCUT_ACTIVE_S_1	Minimum vertical spacing between GCUT and channel	$\geq$	5	nm
GCUT_GATE_EX_1	Minimum horizontal extension of GCUT past GATE	$\geq$	16	nm
GCUT_GATE_S_2	Minimum spacing of GCUT to GATE, when the former is not being used to cut a particular polygon drawn using the latter	$\geq$	16	nm
GCUT_S_3	Minimum vertical spacing between two GCUT layer polygons	$\geq$	40	nm
GCUT_O_1	Overlap of GATE and GCUT in GATE poly direction	$\geq$	5	nm
GCUT_AUX_1	GCUT layer may not exist without the layer GATE_			
GCUT_AUX_2	GCUT layer vertical edge may not lie inside, or coincide with, the GATE layer			
GCUT_AUX_3	GCUT may not interact with channel.			

Table3.8.1 GCUT design rules.

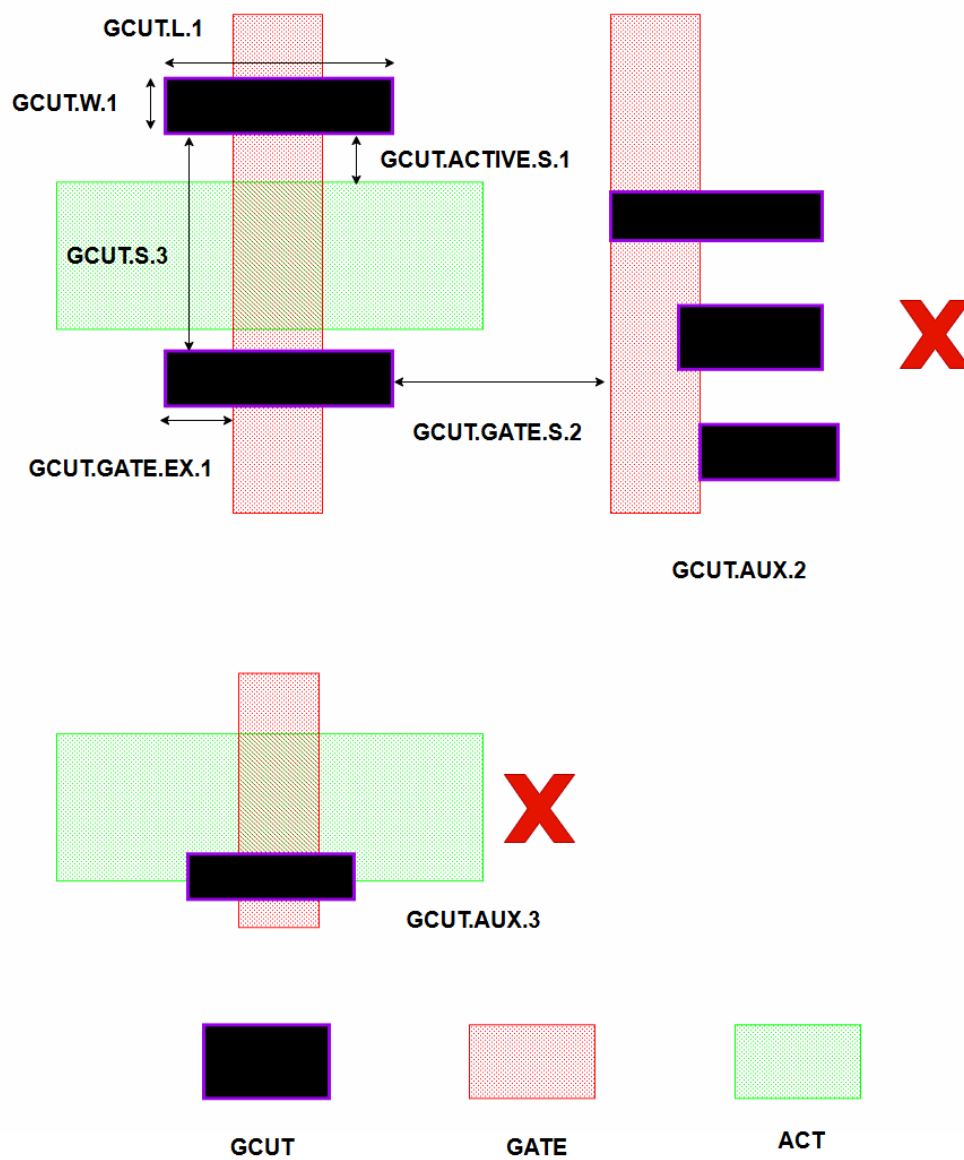


Fig. 3.8.1 Illustration of GCUT design rules.

3.9 CT_PR design rules

Rule number	Description	Opt.	Design value	Unit
CT_PR_W_1	Minimum horizontal width of CT_PR	>=	10	nm
CT_PR_W_2	Minimum vertical width of CT_PR	>=	60	nm
CT_PR_S_1	Minimum horizontal spacing between two CT_PR layer polygons	>=	48	nm
CT_PR_GATE_S_2	Minimum horizontal spacing between CT_PR and GATE	>=	6	nm
CT_PR_ACTIVE_OV_1	Minimum vertical overlap between CT_PR and ACTIVE	>=	20	nm
CT_PR_VCT_PR_OV_2	Minimum vertical overlap between CT_PR and VCT_PR	>=	15	nm
CT_PR_GATE_AUX_1	CT_PR and GATE may not overlap or touch			
CT_PR_ACT_AUX_3	CT_PR may not be entirely outside ACT			

Table3.9.1 CT_PR design rules.

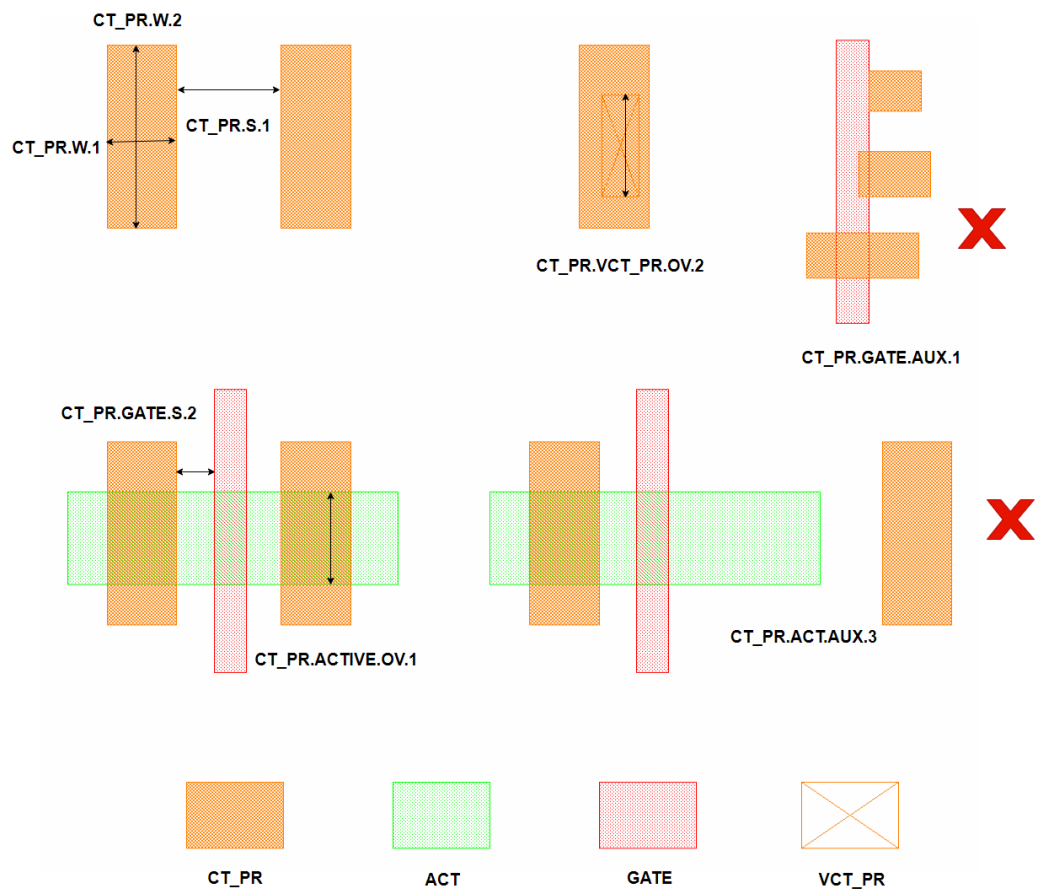


Fig. 3.9.1 Illustration of CT_PR design rules.

3.10 VCT design rules

Rule number	Description	Opt.	Design value	Unit
VCT_W_1	Minimum width of a VCT instance along the length of M0	>=	10	nm
VCT_S_1	Minimum spacing between VCT instances [on the same M0 track on parallel M0 tracks, if they are not aligned with each other on parallel M0 tracks, if they are fully or partially aligned with each other]	>=	20	nm
VCT_S_2	Minimum corner-to-corner spacing between two VCT instances—both with a 5 nm M0 end-cap	>=	23	nm
VCT_S_3	Minimum corner-to-corner spacing between two VCT instances—both without a 5 nm M0 end-cap	>=	30	nm
VCT_S_4	Minimum corner-to-corner spacing between two VCT instances—one with and another without, a 5 nm M0 end-cap	>=	27	nm
VCT_CT_EN_1	Minimum enclosure of VCT by CT on two opposite sides (outside the layer SRAMMRK)	>=	3_5	nm
VCT_M0_EN_1	Minimum enclosure of VCT by M0 on two opposite sides (outside the layer SRAMMRK)	>=	4	nm
VCT_AUX_1	VCT must always interact with M0 and CT			
VCT_M0_AUX_3	VCT must exactly be the same width as M0 along the horizontal direction			

Table3.10.1 VCT design rules.

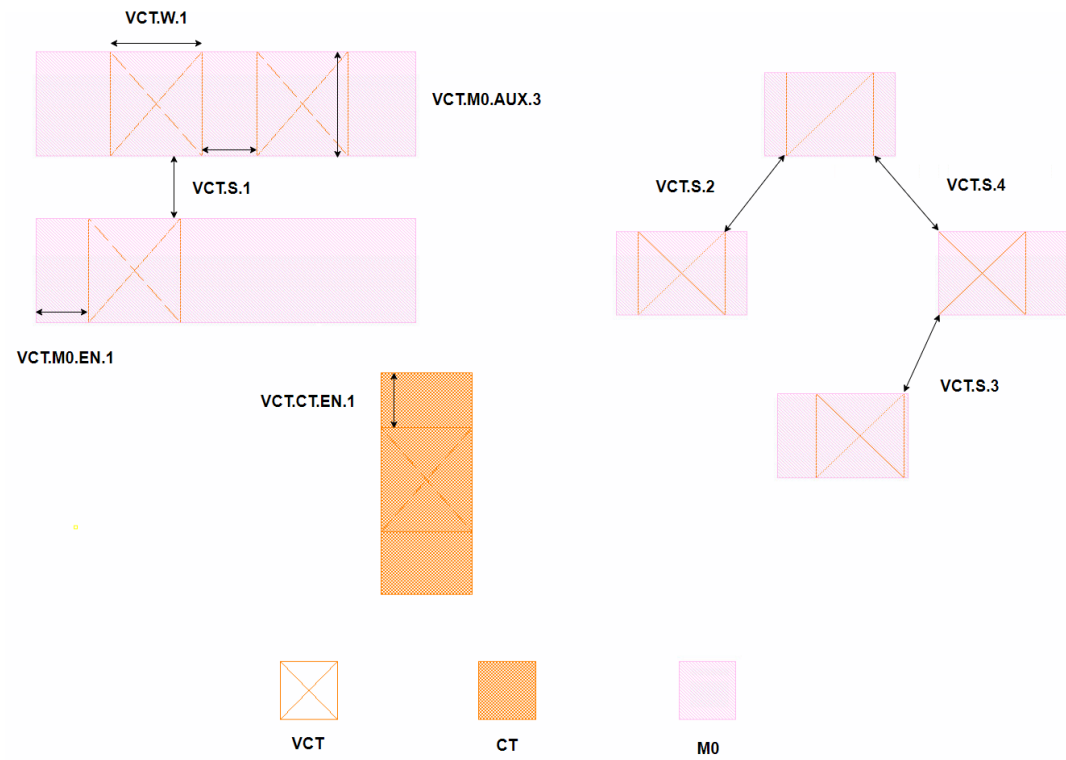


Fig. 3.10.1 Illustration of VCT design rules.

3.11 VCT_PR design rules

Rule number	Description	Opt.	Design value	Unit
VCT_PR_W_1	Minimum width of a VCT_PR instance along the length of M0_PR	>=	19	nm
VCT_PR_S_1	Minimum spacing between VCT_PR instances	>=	48	nm
VCT_PR_S_2	Minimum corner-to-corner spacing between two VCT_PR instances—both with a 5 nm M0_PR end-cap	>=	23	nm
VCT_PR_S_3	Minimum corner-to-corner spacing between two VCT_PR instances—both without a 5 nm M0_PR end-cap	>=	30	nm
VCT_PR_S_4	Minimum corner-to-corner spacing between two VCT_PR instances—one with and another without, a 5 nm M0_PR end-cap	>=	27	nm
VCT_PR_M0_PR_EN_1	Minimum enclosure of VCT_PR by M0_PR on two opposite sides	>=	4	nm
VCT_PR_AUX_1	VCT_PR must always interact with M0_PR and CT_PR			
VCT_PR_M0_PR_AUX_3	VCT_PR must exactly be the same width as M0_PR along the horizontal direction			

Table3.11.1 VCT_PR design rules.

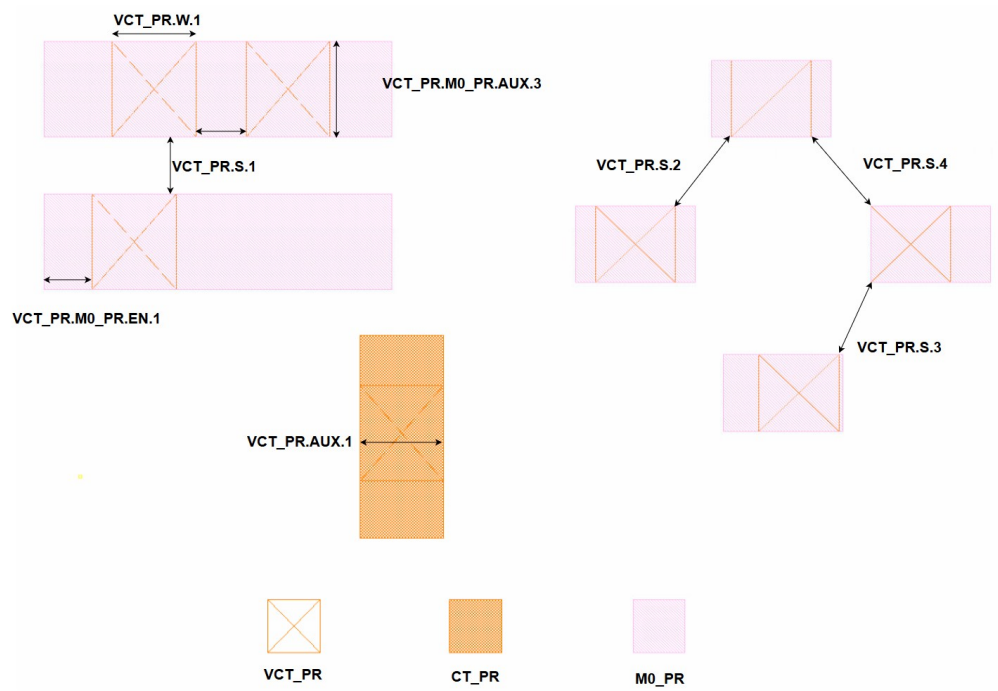


Fig. 3.11.1 Illustration of VCT_PR design rules.

3.12 VCG design rules

Rule number	Description	Opt.	Design value	Unit
VCG_W_1	Minimum width of VCG	>=	14	nm
VCG_GATE_S_1	Minimum horizontal spacing between VCG and GATE (not cut using GCUT) layer polygons	>=	30	nm
VCG_GATE_EN_1	Minimum enclosure of VCG by GATE on two opposite sides	>=	1	nm
VCG_M0_EN_1	Minimum enclosure of VCG by M0 on two opposite sides	>=	4	nm
VCG_VCT_S_1	Minimum corner-to-corner spacing between VCG and VCT	>=	15	nm
VCG_AUX_1	VCG must always interact with M0 and GATE			
VCG_M0_AUX_1	VCG must exactly be the same width as M0 along the direction perpendicular to the M0 length			

Table3.12.1 VCG design rules.

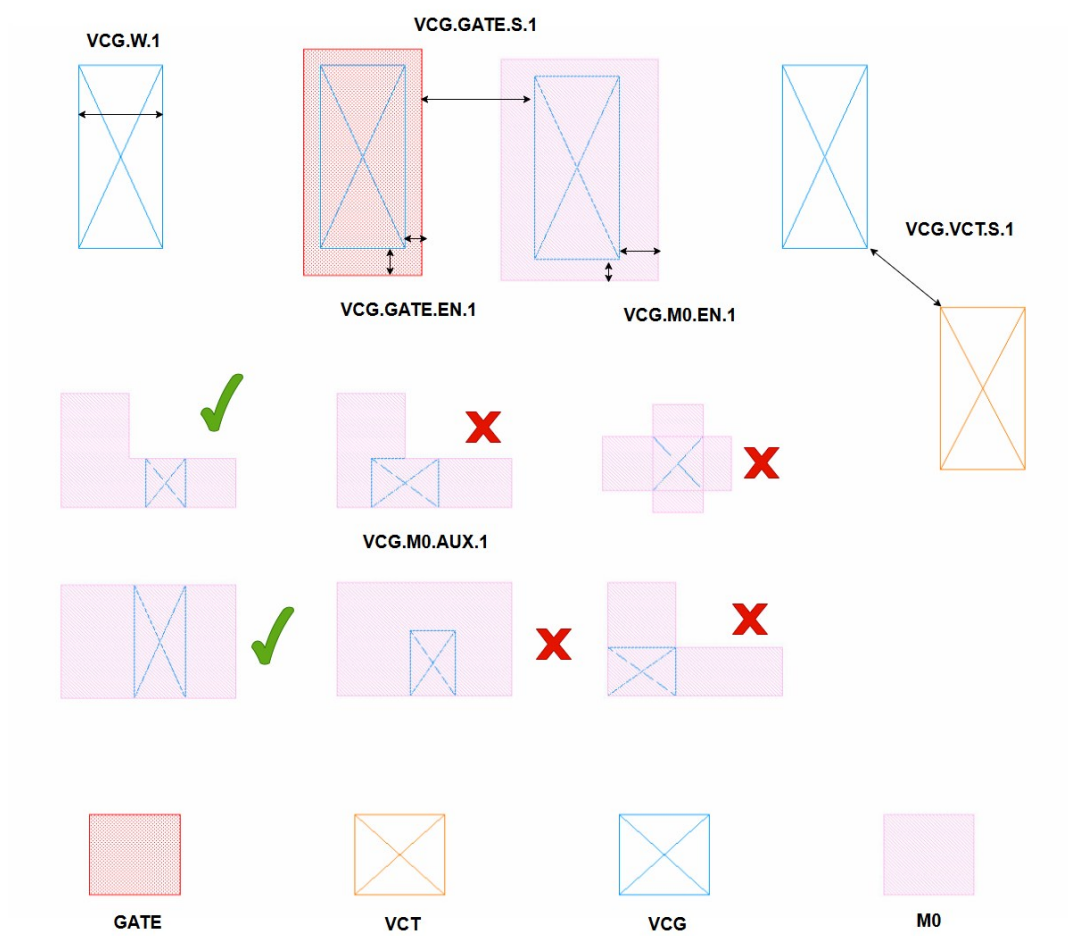


Fig. 3.12.1 Illustration of VCG design rules.

3.13 M0 design rules

Rule number	Description	Opt.	Design value	Unit
M0_W_1	Minimum vertical width of M0	>=	15	nm
M0_W_2	Minimum horizontal width of M0,except MARKS region	>=	25	nm
M0_S_1	Minimum vertical spacing of M0s	>=	11	nm
M0_M0_PR_S_1	Minimum vertical spacing between M0 and M0_PR	>=	11	nm
M0_S_2	Minimum horizontal (TtT) spacing between two M0 layers on the same track	>=	18	nm
M0_A_1	Area of M0	>=	375	nm2
M0_S_3	Minimum corner-to-corner spacing between two M0 polygons	>=	15	nm
M0_AUX_1	M0 must not bend			

Table3.13.1 M0 design rules.

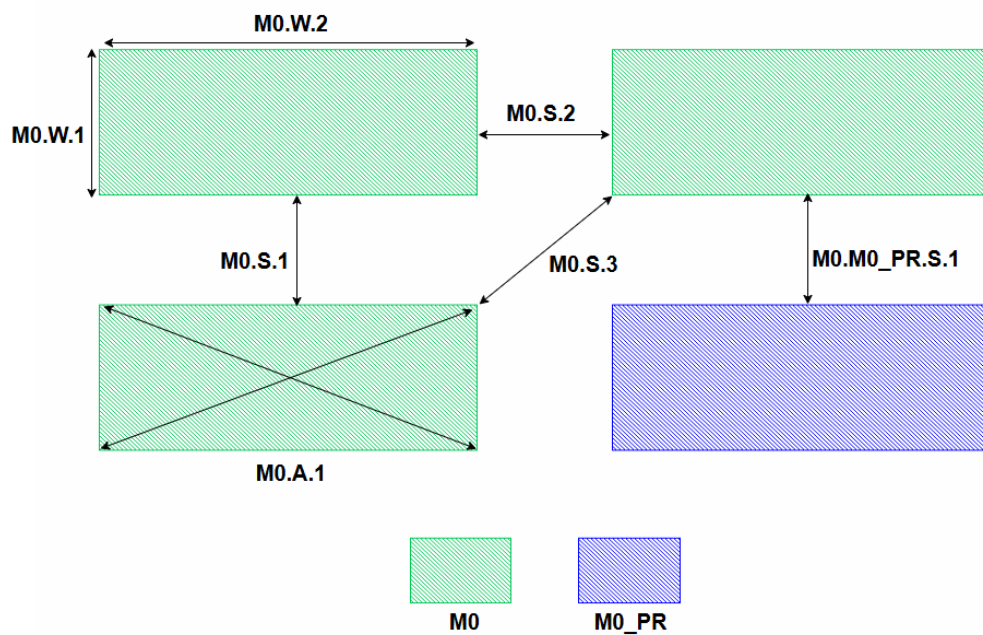


Fig. 3.13.1 Illustration of M0 design rules.

3.14 M0_PR design rules

Rule number	Description	Opt.	Design value	Unit
M0_PR_W_1	Minimum vertical width of M0_PR	>=	38	nm
M0_PR_W_2	Minimum horizontal width of M0_PR, except MARKS region	>=	90	nm
M0_PR_S_1	Minimum vertical spacing of M0_PR s,	>=	98	nm
M0_PR_A_1	Area of M0_PR	>=	1710	nm2
M0_PR_S_2	Minimum horizontal (TtT) spacing between two M0_PR layers on the same track	>=	60	nm
M0_PR_S_3	Minimum corner-to-corner spacing between two M0_PR polygons	>=	50	nm
M0_PR_AUX_1	M0_PR must not bend			

Table3.14.1 M0_PR design rules.

3.15 V0 design rules

Rule number	Description	Opt.	Design value	Unit
V0_W_1	Minimum width of V0(except SealRing)	>=	15	nm
V0_S_1	Minimum corner to corner spacing between V0 instances	>=	14	nm
V0_M0_EN_1	V0 must be covered by M0 or M0_PR			
V0_M0_EN_2	V0 enclosure by M0 or M0_PR on horizontal direction	>=	5	nm
V0_M0_AUX_1	V0 must exactly be the same vertical width with M0			
V0_M1_AUX_2	V0 must exactly be the same horizontal width with M1			
V0_AUX_3	V0 must interact with M0 and M1			
V0_M1_EN_1	V0 enclosure by M1 on two opposite sides	>=	5	nm

Table3.15.1 V0 design rules.

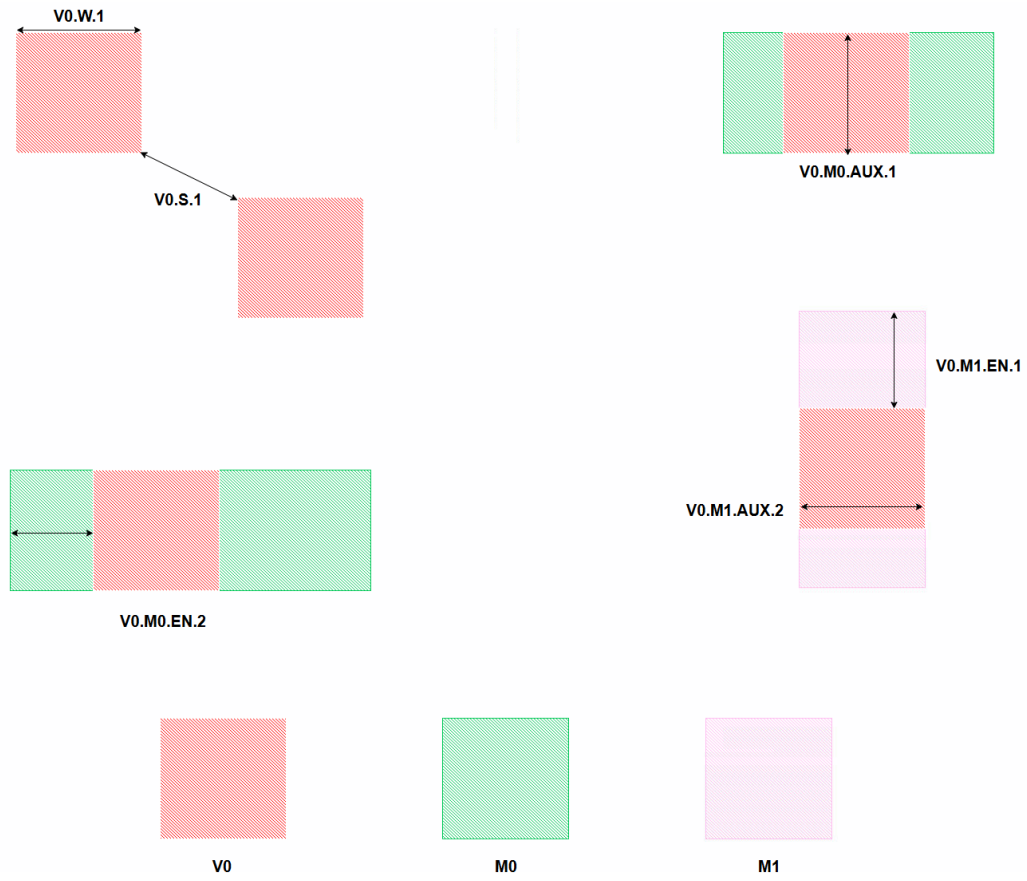


Fig. 3.15.1 Illustration of V0 design rules.

3.16 M1 design rules

Rule number	Description	Opt.	Design value	Unit
M1_W_1	Minimum vertical width of M1	>=	28	nm
M1_W_2	Minimum horizontal width of M1	>=	21	nm
M1_S_1	Minimum horizontal spacing of M1s,	>=	13	nm
M1_S_2	Minimum vertical (TtT) spacing between two M1 layers on the same track	>=	27	nm
M1_A_1	Area of M1	>=	504	nm ²
M1_S_3	Minimum corner-to-corner spacing between two M1 polygons	>=	24	nm
M1_AUX_1	M1 must not bend			

Table3.16.1 M1 design rules.

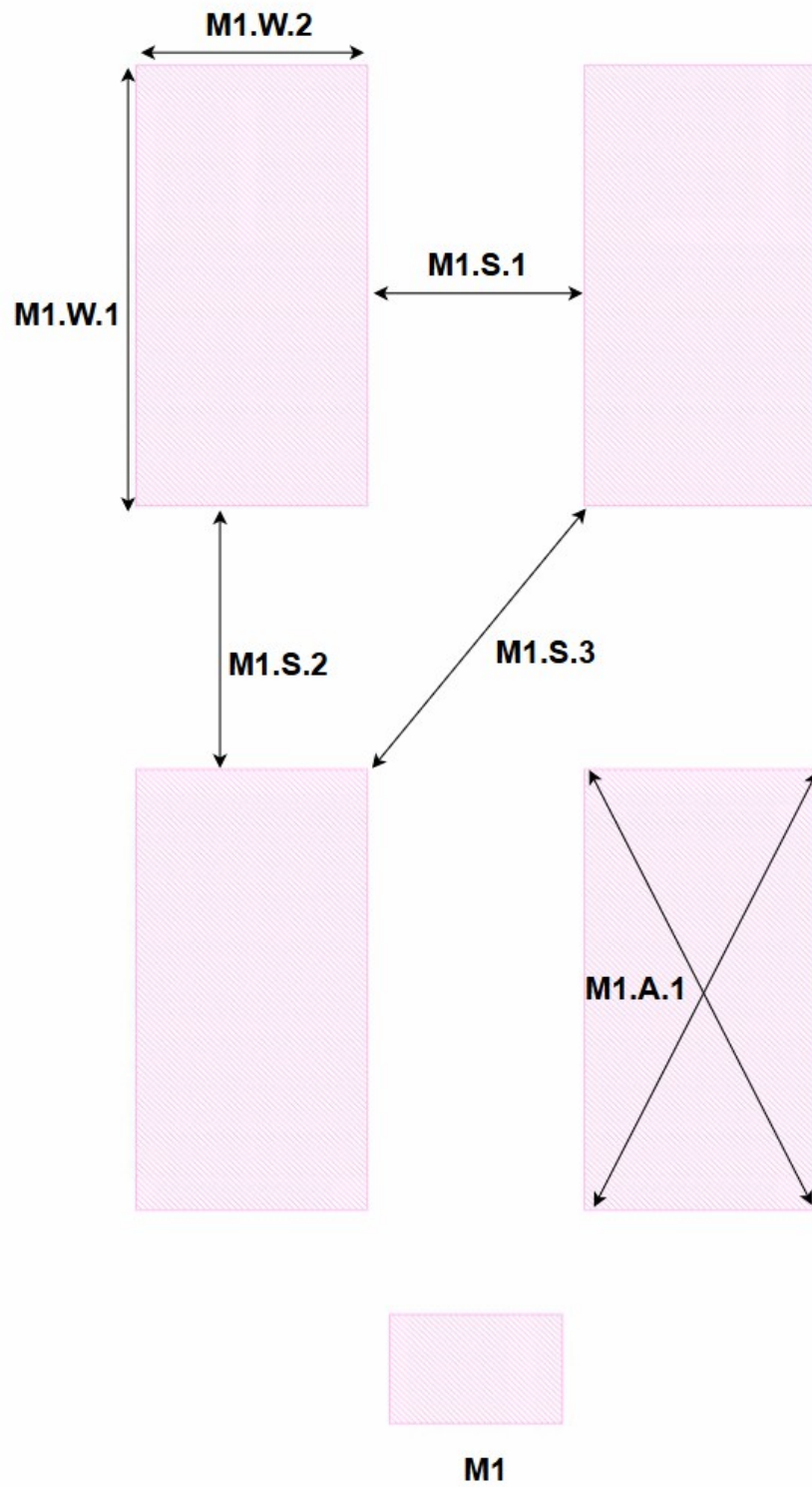


Fig. 3.16.1 Illustration of M1 design rules.

3.17 V1 design rules

Rule number	Description	Opt.	Design value	Unit
V1_W_1	Minimum width	>=	21	nm
V1_S_1	Minimum corner to corner spacing between V1 instances	>=	15	nm
V1_M1_AUX_1	V1 must exactly be the same horizontal width with M1			
V1_M2_AUX_2	V1 must exactly be the same vertical width with M2			
V1_AUX_3	V1 must interact with M1 and M2			
V1_M1_EN_2	V1 enclosure by M1	>=	5	nm
V1_M2_EN_1	V1 enclosure by M2	>=	5	nm

Table3.17.1 V1 design rules.

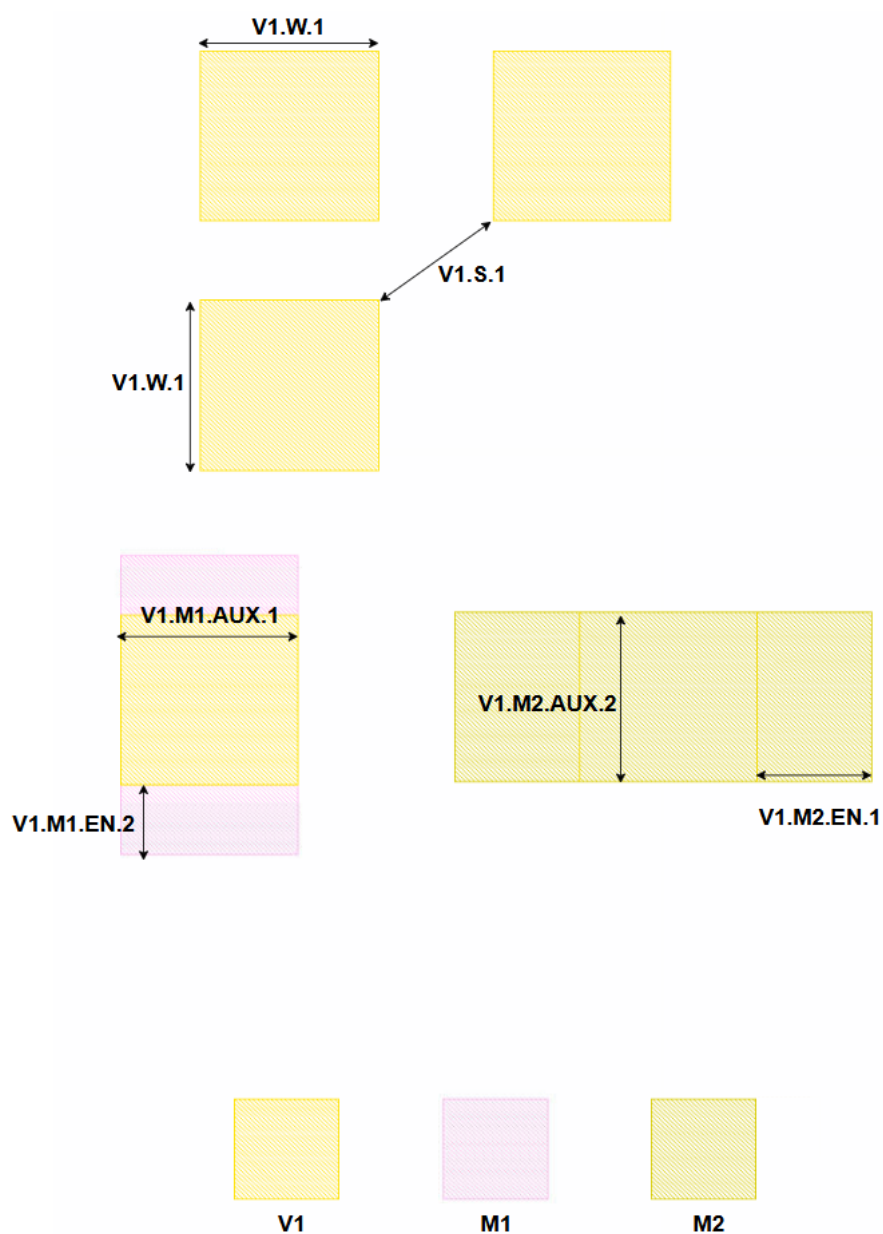


Fig. 3.17.1 Illustration of V1 design rules.

3.18 M2 design rules

Rule number	Description	Opt.	Design value	Unit
M2_W_1	Minimum vertical width	>=	21	nm
M2_S_1	Minimum vertical spacing	>=	13	nm
M2_S_2	Minimum horizontal (TtT) spacing between two M2 layers on the same track	>=	20	nm
M2_S_3	Minimum corner-to-corner spacing between two M2 polygons	>=	16	nm
M2_AUX_1	M2 must not bend			

Table3.18.1 M2 design rules.

3.19 V2 design rules

Rule number	Description	Opt.	Design value	Unit
V2_W_1	Minimum width	>=	21	nm
V2_S_1	Minimum corner to corner spacing between V2 instances	>=	15	nm
V2_M2_AUX_1	V2 must exactly be the same vertical width with M2			
V2_M3_AUX_2	V2 must exactly be the same horizontal width with M3			
V2_AUX_3	V2 must interact with M2 and M3			
V2_M2_EN_2	V2 enclosure by M2	>=	5	nm
V2_M3_EN_1	V2 enclosure by M3	>=	5	nm

Table3.19.1 V2 design rules.

3.20 M3 design rules

Rule number	Description	Opt.	Design value	Unit
M3_W_1	Minimum horizontal width	>=	21	nm
M3_S_1	Minimum horizontal spacing	>=	13	nm
M3_S_2	Minimum horizontal (TtT) spacing between two M3 layers on the same track	>=	20	nm
M3_S_3	Minimum corner-to-corner spacing between two M3 polygons	>=	16	nm
M3_AUX_1	M3 must not bend			

Table3.20.1 M3 design rules.

3.21 V3 design rules

Rule number	Description	Opt.	Design value	Unit
V3_W_1	Minimum width	>=	21	nm
V3_S_1	Minimum corner to corner spacing between V3 instances	>=	15	nm
V3_M3_AUX_1	V3 must exactly be the same horizontal width with M3			
V3_M4_AUX_2	V3 must exactly be the same vertical width with M4			
V3_AUX_3	V3 must interact with M3 and M4			
V3_M3_EN_2	V3 enclosure by M3	>=	5	nm
V3_M4_EN_1	V3 enclosure by M4	>=	5	nm

Table3.21.1 V3 design rules.

3.22 M4 design rules

Rule number	Description	Opt.	Design value	Unit
M4_W_1	Minimum vertical width	>=	30	nm
M4_S_1	Minimum vertical spacing	>=	25	nm
M4_S_2	Minimum horizontal (TtT) spacing between two M4 layers on the same track	>=	32	nm
M4_S_3	Minimum corner-to-corner spacing between two M4 polygons	>=	28	nm
M4_AUX_1	M4 must not bend			

Table3.22.1 M4 design rules.

3.23 V4 design rules

Rule number	Description	Opt.	Design value	Unit
V4_W_1	Minimum width	>=	30	nm
V4_S_1	Minimum corner to corner spacing between V4 instances	>=	25	nm
V4_M4_AUX_1	V4 must exactly be the same vertical width with M4			
V4_M5_AUX_2	V4 must exactly be the same horizontal width with M5			
V4_AUX_3	V4 must interact with M4 and M5			
V4_M4_EN_2	V4 enclosure by M4	>=	5	nm
V4_M5_EN_1	V4 enclosure by M5	>=	5	nm

Table3.23.1 V4 design rules.

3.24 M5 design rules

Rule number	Description	Opt.	Design value	Unit
M5_W_1	Minimum horizontal width	>=	30	nm
M5_S_1	Minimum horizontal spacing	>=	25	nm
M5_S_2	Minimum horizontal (TtT) spacing between two M5 layers on the same track	>=	32	nm
M5_S_3	Minimum corner-to-corner spacing between two M5 polygons	>=	28	nm
M5_AUX_1	M5 must not bend			

Table3.24.1 M5 design rules.

3.25 V5 design rules

Rule number	Description	Opt.	Design value	Unit
V5_W_1	Minimum width	>=	30	nm
V5_S_1	Minimum corner to corner spacing between V5 instances	>=	25	nm
V5_M5_AUX_1	V5 must exactly be the same horizontal width with M5			
V5_M6_AUX_2	V5 must exactly be the same vertical width with M6			
V5_AUX_3	V5 must interact with M5 and M6			
V5_M5_EN_2	V5 enclosure by M5	>=	5	nm
V5_M6_EN_1	V5 enclosure by M6	>=	5	nm

Table3.25.1 V5 design rules.

3.26 M6 design rules

Rule number	Description	Opt.	Design value	Unit
M6_W_1	Minimum vertical width	>=	30	nm
M6_S_1	Minimum vertical spacing	>=	25	nm
M6_S_2	Minimum horizontal (TtT) spacing between two M6 layers on the same track	>=	32	nm
M6_S_3	Minimum corner-to-corner spacing between two M6 polygons	>=	28	nm
M6_AUX_1	M6 must not bend			

Table3.26.1 M6 design rules.

3.27 V6 design rules

Rule number	Description	Opt.	Design value	Unit
V6_W_1	Minimum width	>=	30	nm
V6_S_1	Minimum corner to corner spacing between V6 instances	>=	25	nm
V6_M6_AUX_1	V6 must exactly be the same vertical width with M6			
V6_M7_AUX_2	V6 must exactly be the same horizontal width with M7			
V6_AUX_3	V6 must interact with M6 and M7			
V6_M6_EN_2	V6 enclosure by M6	>=	5	nm
V6_M7_EN_1	V6 enclosure by M7	>=	5	nm

Table3.27.1 V6 design rules.

3.28 M7 design rules

Rule number	Description	Opt.	Design value	Unit
M7_W_1	Minimum horizontal width	>=	35	nm
M7_S_1	Minimum horizontal spacing	>=	25	nm
M7_S_2	Minimum horizontal (TtT) spacing between two M7 layers on the same track	>=	32	nm
M7_S_3	Minimum corner-to-corner spacing between two M7 polygons	>=	28	nm
M7_AUX_1	M7 must not bend			

Table3.28.1 M7 design rules.

3.29 V7 design rules

Rule number	Description	Opt.	Design value	Unit
V7_W_1	Minimum width	>=	35	nm
V7_S_1	Minimum corner to corner spacing between V7 instances	>=	25	nm
V7_M7_AUX_1	V7 must exactly be the same horizontal width with M7			
V7_M8_AUX_2	V7 must exactly be the same vertical width with M8			
V7_AUX_3	V7 must interact with M7 and M8			
V7_M7_EN_2	V7 enclosure by M7	>=	5	nm
V7_M8_EN_1	V7 enclosure by M8	>=	5	nm

Table3.29.1 V7 design rules.

3.30 M8 design rules

Rule number	Description	Opt.	Design value	Unit
M8_W_1	Minimum vertical width	>=	35	nm
M8_S_1	Minimum vertical spacing	>=	25	nm
M8_S_2	Minimum horizontal (TtT) spacing between two M8 layers on the same track	>=	32	nm
M8_S_3	Minimum corner-to-corner spacing between two M8 polygons	>=	28	nm
M8_AUX_1	M8 must not bend			

Table3.30.1 M8 design rules.

3.31 V8 design rules

Rule number	Description	Opt.	Design value	Unit
V8_W_1	Minimum width	>=	35	nm
V8_S_1	Minimum corner to corner spacing between V8 instances	>=	25	nm
V8_M8_AUX_1	V8 must exactly be the same vertical width with M8			
V8_M9_AUX_2	V8 must exactly be the same horizontal width with M9			
V8_AUX_3	V8 must interact with M8 and M9			
V8_M8_EN_2	V8 enclosure by M8	>=	5	nm
V8_M9_EN_1	V8 enclosure by M9	>=	5	nm

Table3.31.1 V8 design rules.

3.32 M9 design rules

Rule number	Description	Opt.	Design value	Unit
M9_W_1	Minimum horizontal width	>=	35	nm
M9_S_1	Minimum horizontal spacing	>=	25	nm
M9_S_2	Minimum horizontal (TtT) spacing between two M9 layers on the same track	>=	32	nm
M9_S_3	Minimum corner-to-corner spacing between two M9 polygons	>=	28	nm
M9_AUX_1	M9 must not bend			

Table3.32.1 M9 design rules.

3.33 V9 design rules

Rule number	Description	Opt.	Design value	Unit
V9_W_1	Minimum width	>=	35	nm
V9_S_1	Minimum corner to corner spacing between V9 instances	>=	25	nm
V9_M9_AUX_1	V9 must exactly be the same horizontal width with M9			
V9_M10_AUX_2	V9 must exactly be the same vertical width with M10			
V9_AUX_3	V9 must interact with M9 and M10			

Table3.33.1 V9 design rules.

3.34 M10 design rules

Rule number	Description	Opt.	Design value	Unit
M10_W_1	Minimum vertical width	>=	50	nm
M10_S_1	Minimum vertical spacing	>=	40	nm
M10_S_2	Minimum horizontal (TtT) spacing between two M10 layers on the same track	>=	50	nm
M10_S_3	Minimum corner-to-corner spacing between two M10 polygons	>=	45	nm
M10_AUX_1	M10 must not bend			

Table3.34.1 M10 design rules.

3.35 V10 design rules

Rule number	Description	Opt.	Design value	Unit
V10_W_1	Minimum width	>=	50	nm
M10_W_1	Minimum vertical width	>=	50	nm
M10_S_1	Minimum vertical spacing	>=	40	nm
M10_S_2	Minimum horizontal (TtT) spacing between two M10 layers on the same track	>=	50	nm
M10_S_3	Minimum corner-to-corner spacing between two M10 polygons	>=	45	nm
M10_AUX_1	M10 must not bend			

Table3.35.1 V10 design rules.

3.36 M11 design rules

Rule number	Description	Opt.	Design value	Unit
M11_W_1	Minimum horizontal width	>=	50	nm
M11_S_1	Minimum horizontal spacing	>=	40	nm
M11_S_2	Minimum horizontal (TtT) spacing between two M11 layers on the same track	>=	50	nm
M11_S_3	Minimum corner-to-corner spacing between two M11 polygons	>=	45	nm
M11_AUX_1	M11 must not bend			

Table3.36.1 M11 design rules.

3.37 V11 design rules

Rule number	Description	Opt.	Design value	Unit
V11_W_1	Minimum width	>=	50	nm
V11_S_1	Minimum corner to corner spacing between V11 instances	>=	40	nm
V11_M11_AUX_1	V11 must exactly be the same horizontal width with M11			
V11_M12_AUX_2	V11 must exactly be the same vertical width with M12			
V11_AUX_3	V11 must interact with M11 and M12			

Table3.37.1 V11 design rules.

3.38 M12 design rules

Rule number	Description	Opt.	Design value	Unit
M12_W_1	Minimum vertical width	>=	60	nm
M12_S_1	Minimum vertical spacing	>=	40	nm
M12_S_2	Minimum horizontal (TtT) spacing between two M12 layers on the same track	>=	50	nm
M12_S_3	Minimum corner-to-corner spacing between two M12 polygons	>=	45	nm
M12_AUX_1	M12 must not bend			

Table3.38.1 M12 design rules.

3.39 V12 design rules

Rule number	Description	Opt.	Design value	Unit
V12_W_1	Minimum width	>=	60	nm
V12_S_1	Minimum corner to corner spacing between V12 instances	>=	40	nm
V12_M12_AUX_1	V12 must exactly be the same vertical width with M12			
V12_M13_AUX_2	V12 must exactly be the same horizontal width with M13			
V12_AUX_3	V12 must interact with M12 and M13			

Table3.39.1 V12 design rules.

3.40 M13 design rules

Rule number	Description	Opt.	Design value	Unit
M13_W_1	Minimum horizontal width	>=	60	nm
M13_S_1	Minimum horizontal spacing	>=	40	nm
M13_S_2	Minimum horizontal (TtT) spacing between two M13 layers on the same track	>=	50	nm
M13_S_3	Minimum corner-to-corner spacing between two M13 polygons	>=	45	nm
M13_AUX_1	M13 must not bend			

Table3.40.1 M13 design rules.